P2 Chapter 9: Differentiation

The Chain Rule

The Product Rule

As mentioned previously, the product rule is used, unsurprisingly, when we have a **product** of two functions.

✏ The product rule:
If $y = uv$ then $\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

This is quite easy to remember. Differentiate one of the things but leave the other. Then do the other way round. Then add!
Since addition is commutative, it doesn't matter which way round we do it.

If $y = x^2 \sin x$, determine $\frac{dy}{dx}$

$$u = x^2 \qquad v = \sin x$$
$$\frac{du}{dx} = 2x \qquad \frac{dv}{dx} = \cos x$$
$$\frac{dy}{dx} = x^2 \cos x + 2x \sin x$$

Write out $u = \cdots, v = \cdots$ then differentiate each.

Sub into product rule.

Fro Tip: While with the chain rule I encourage you to put the bits together without writing out $u =$, in contrast, with the product rule I do encourage you to write out in full as per the left. This is because each of u and v can be more complicated to differentiate – it thus helps you avoid trying to do too much at once.

If $y = xe^{2x}$, determine the coordinates of the turning point.

$$u = x \qquad v = e^{2x}$$
$$\frac{du}{dx} = 1 \qquad \frac{dv}{dx} = 2e^{2x}$$
$$\frac{dy}{dx} = 2xe^{2x} + e^{2x} = 0$$

$e^{2x}(2x+1) = 0$
But $e^{2x} \neq 0$

$$\therefore 2x + 1 = 0 \rightarrow x = -\frac{1}{2}$$
$$y = -\frac{1}{2}e^{-1} = -\frac{1}{2e} \rightarrow \left(-\frac{1}{2}, -\frac{1}{2e}\right)$$

It's typical to factorise when finding a turning point.

The fact that exponential terms can never be 0 comes up frequently in exams.

Product + Chain Rule Examples

[Textbook] If $y = e^{4x} \sin^2 3x$, show that $\frac{dy}{dx} = e^{4x} \sin 3x \, (A \cos 3x + B \sin 3x)$, where A and B are constants to be determined.

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[Textbook] Given that $f(x) = x^2\sqrt{3x - 1}$, find $f'(x)$

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$$\boldsymbol{u = e^{4x} \qquad v = (\sin 3x)^2}$$

$$\boldsymbol{\frac{du}{dx} = 4e^{4x} \qquad \frac{dv}{dx} = 2\sin 3x \times 3\cos 3x = 6\sin 3x\cos 3x}$$

$$\boldsymbol{\frac{dy}{dx} = 6e^{4x}\sin 3x\cos 3x + 4e^{4x}(\sin 3x)^2}$$

$$\boldsymbol{= e^{4x}\sin 3x\,(6\cos 3x + 4\sin 3x)}$$

[Textbook] Given that $f(x) = x^2\sqrt{3x-1}$, find $f'(x)$

$$\boldsymbol{u = x^2 \qquad v = (3x-1)^{\frac{1}{2}}}$$

$$\boldsymbol{\frac{du}{dx} = 2x \qquad \frac{dv}{dx} = \frac{1}{2}(3x-1)^{-\frac{1}{2}} \times 3 = \frac{3}{2}(3x-1)^{-\frac{1}{2}}}$$

$$\boldsymbol{\frac{dy}{dx} = \frac{3}{2}x^2(3x-1)^{-\frac{1}{2}} + 2x(3x-1)^{\frac{1}{2}}}$$

$$\boldsymbol{= \frac{1}{2}x(3x-1)^{-\frac{1}{2}}[3x + 4(3x-1)]}$$

$$\boldsymbol{= \frac{1}{2}x(3x-1)^{-\frac{1}{2}}(15x - 4)}$$

$$\boldsymbol{= \frac{x(15x-4)}{2\sqrt{3x-1}}}$$

This is when the manipulation is a little tricky. Remember to:

(a) Factor out the smallest power, in this case $-\frac{1}{2}$. Since a power of $\frac{1}{2}$ is 1 more than $-\frac{1}{2}$, we have an extra $(3x-1)$

(b) Factor out the fraction. $\frac{1}{2}$ goes into both $\frac{3}{2}$ and 2.

This last step is optional. Try to avoid negative powers, and $\sqrt{}$ is tidier.

Test Your Understanding

Edexcel C3 Jan 2012 Q1a

Differentiate with respect to x, giving your answer in its simplest form,

(*a*) $x^2 \ln(3x)$, **(4)**

?

Edexcel C3 June 2013 Q5(c)

$$\frac{dy}{dx} = \frac{1}{6x(x-1)^{\frac{1}{2}}}$$

Find $\frac{d^2y}{dx^2}$, simplifying your answer.

Examiner's Report: "Perhaps the most challenging question in the paper". And this is on a paper described as having "a level of demand higher than any previous C3 paper". So it's DAMN FRIKKIN' HARD. 😐😐

?

Test Your Understanding

Edexcel C3 Jan 2012 Q1a

Differentiate with respect to x, giving your answer in its simplest form,

(*a*) $x^2 \ln(3x)$, **(4)**

(a)	$\frac{d}{dx}(\ln(3x)) \to \frac{B}{x}$ for any constant B	M1
Applying vu'+uv' ,	$\ln(3x) \times 2x + x$	M1, A1 A1
		(4)

Edexcel C3 June 2013 Q5(c)

$$\frac{dy}{dx} = \frac{1}{6x(x-1)^{\frac{1}{2}}}$$

Find $\frac{d^2y}{dx^2}$, simplifying your answer.

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$$\frac{dy}{dx} = \frac{1}{6}x^{-1}(x-1)^{-\frac{1}{2}}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{6}x^{-2}(x-1)^{-\frac{1}{2}} - \frac{1}{12}x^{-1}(x-1)^{-\frac{3}{2}}$$

$$= -\frac{1}{12}x^{-2}(x-1)^{-\frac{3}{2}}(2(x-1)+x)$$

$$= \frac{2-3x}{12x^2(x-1)^{\frac{3}{2}}}$$

Exercise 9.4

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Homework Exercise

1 Differentiate:

a $x(1 + 3x)^5$ **b** $2x(1 + 3x^2)^3$ **c** $x^3(2x + 6)^4$ **d** $3x^2(5x - 1)^{-1}$

2 Differentiate:

a $e^{-2x}(2x - 1)^5$ **b** $\sin 2x \cos 3x$ **c** $e^x \sin x$ **d** $\sin(5x) \ln(\cos x)$

3 **a** Find the value of $\frac{dy}{dx}$ at the point (1, 8) on the curve with equation $y = x^2(3x - 1)^3$.

b Find the value of $\frac{dy}{dx}$ at the point (4, 36) on the curve with equation $y = 3x(2x + 1)^{\frac{1}{2}}$.

c Find the value of $\frac{dy}{dx}$ at the point $\left(2, \frac{1}{5}\right)$ on the curve with equation $y = (x - 1)(2x + 1)^{-1}$.

4 Find the stationary points of the curve C with the equation $y = (x - 2)^2(2x + 3)$.

5 A curve C has equation $y = \left(x - \frac{\pi}{2}\right)^5 \sin 2x$, $0 < x < \pi$. Find the gradient of the curve at the point with x-coordinate $\frac{\pi}{4}$

6 A curve C has equation $y = x^2 \cos(x^2)$. Find the equation of the tangent to the curve C at the point $P\left(\frac{\sqrt{\pi}}{2}, \frac{\pi\sqrt{2}}{8}\right)$ in the form $ax + by + c = 0$ where a, b and c are exact constants. **(7 marks)**

Homework Exercise

7 Given that $y = 3x^2(5x - 3)^3$, show that

$$\frac{dy}{dx} = Ax(5x - 3)^n(Bx + C)$$

where n, A, B and C are constants to be determined. **(4 marks)**

8 A curve C has equation $y = (x + 3)^2 e^{3x}$.

a Find $\frac{dy}{dx}$, using the product rule for differentiation. **(3 marks)**

b Find the gradient of C at the point where $x = 2$. **(3 marks)**

9 Differentiate with respect to x:

a $(2\sin x - 3\cos x)\ln 3x$ **(3 marks)**

b $x^4 e^{7x-3}$ **(3 marks)**

10 Find the value of $\frac{dy}{dx}$ at the point where $x = 1$ on the curve with equation

$$y = x^5\sqrt{10x + 6}$$ **(6 marks)**

Challenge

Find $\frac{dy}{dx}$ for the following functions.

a $y = e^x \sin^2 x \cos x$ **b** $y = x(4x - 3)^6(1 - 4x)^9$

Homework Answers

1 **a** $(3x+1)^4(18x+1)$ **b** $2(3x^2+1)^2(21x^2+1)$
c $16x^2(x+3)^3(7x+9)$ **d** $3x(5x-2)(5x-1)^{-2}$

2 **a** $-4(x-3)(2x-1)^4\,e^{-2x}$
b $2\cos 2x\cos 3x - 3\sin 2x\sin 3x$
c $e^x(\sin x+\cos x)$ **d** $5\cos 5x\ln(\cos x) - \tan x\sin 5x$

3 **a** 52 **b** 13 **c** $\frac{3}{25}$

4 $(2, 0), \left(-\frac{1}{3}, \frac{343}{27}\right)$ **5** $\dfrac{5\pi^4}{256}$

6 $\sqrt{2\pi}\,(\pi-4)x + 8y - \pi\sqrt{2}\left(\dfrac{\pi-2}{2}\right) = 0$

7 $6x(5x-3)^3 + 3x^2[3(5x-3)^2(5)] = 6x(5x-3)^3 + 45x^2(5x-3)^2$
$= 3x(5x-3)^2(2(5x-3)+15x) = 3x(5x-3)^2(10x-6+15x)$
$= 3x(5x-3)^2(25x-6) \Rightarrow n = 2, A = 3, B = 25, C = -6$

8 **a** $(x+3)(3x+11)e^{3x}$ **b** $85e^6$

9 **a** $(3\sin x + 2\cos x)\ln(3x) + \dfrac{2\sin x - 3\cos x}{x}$
b $x^3(7x+4)e^{7x-3}$

10 21.25

Challenge

a $-e^x\sin x\,(\sin^2 x - \cos x\sin x - 2\cos^2 x)$
b $-(4x-3)^5(4x-1)^8(256x^2-148x+3)$